Meibomianitis and Associated Skin Disorders

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SOAP Format

- S = Subjective
  - AKA symptoms
  - Everything the patient says
- O = Objective
  - AKA signs
  - All the doctors observes and all the test results
- A = Assessment
  - AKA diagnosis (or diagnoses)
  - What is wrong with the patient
- P = plan
  - For each diagnosis, what is going to be done to help the patient

14/14 up/down

Meibomian Gland Dysfunction

- Not considered an infectious entity
- Altered meibomian gland metabolism
- → secretions become thickened and block the gland opening
- → stagnant material becomes growth medium for bacteria
- → bacteria produce lipase and the released fatty acids trigger inflammation

Meibomian Gland Dysfunction

Often associated with (acne) rosacea

- **Etiology**
  - **Chronic inflammatory disorder involving skin and eye**
  - **Sebaceous glands**

- **Epidemiology** EU - common
  - **"Celtic", northern ancestry (genetic component)**
  - **Adults 30-60 years of age**
  - **> 30% have ocular problems**

Rosacea

- Redness (erythema) in the “facial flush area”



Rosacea

- **Papules and pustules in the "flush areas"**

red leosen

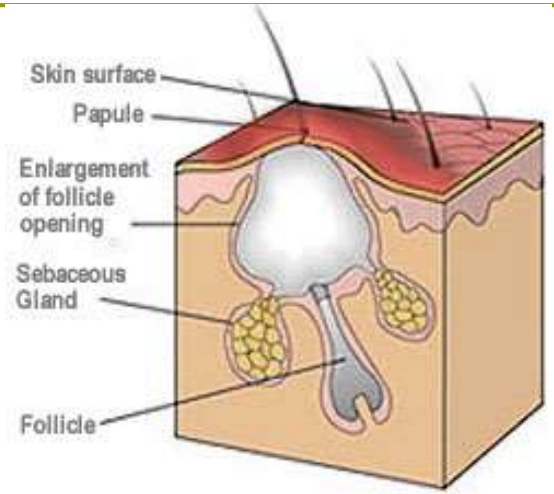




Rosacea

Papule

- **Red and inflamed lesion on the skin's surface**
- **Contains no pus or white head**
- **High break in the follicle wall**
- **Cellular debris and bacteria spill into the dermis.**
- **Often progresses to a pustule**



Papule

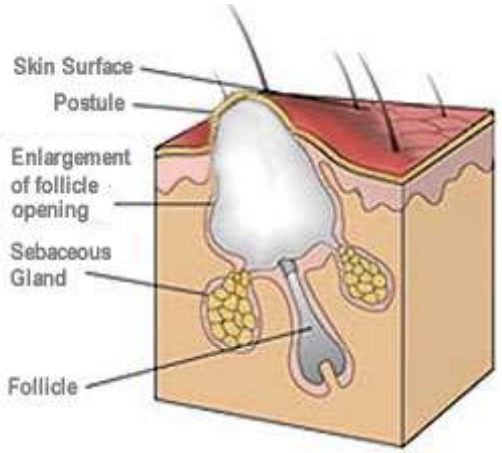


like a "pimple"

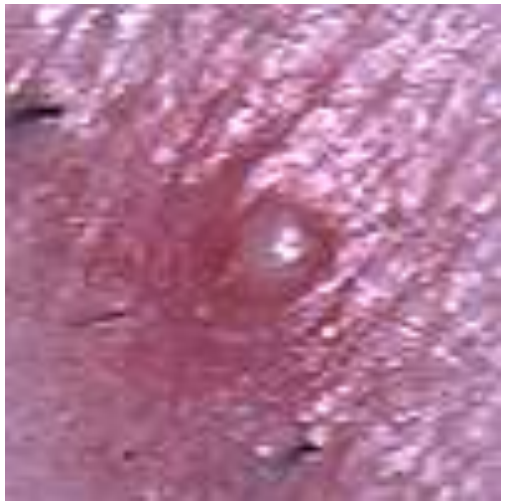
Rosacea

Pustule

- A typical ‘pimple’
- Red and inflamed with a white head
- Sometimes brownish core can be seen as a cap on the pus head
- Pustules follow papules after white blood cells invade
- White cap on a pustule is pus, sebaceous matter and cellular debris



Pustule



Rosacea

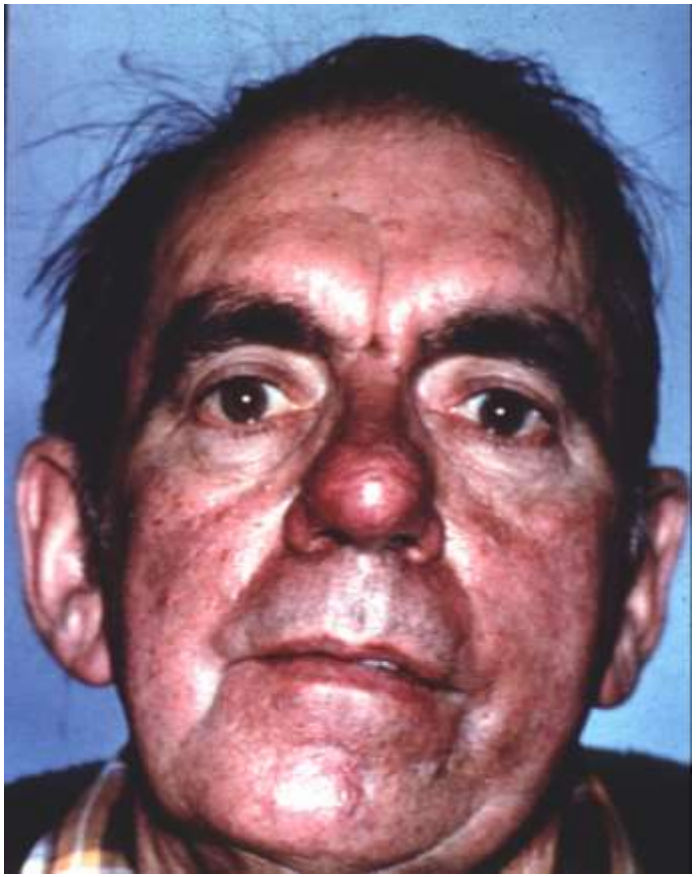
- **Telangiectasias - visible (dilated) small blood vessels on nose or cheeks**



Rosacea

- **Rhinophyma - red, bullous nose (sebaceous gland hypertrophy)**

getting bigger and bigger swelling the nose



- **W.C. Fields 1880 – 1946**

"famous for RHINOPHYMA"
not necessery bigger nose



Rosacea

Treatment of facial rosacea

- Doxycycline 100 mg bid x 4-6 weeks antibaotic
- Topical metrodiazole gel 0.75% bid x 5-8 weeks





Rosacea

Treatment of Facial Rosacea

- Dietary restrictions that trigger reaction
  - well-known rosacea tripwires:
    - hot spices, alcohol, heated beverages
  - lesser-known rosacea tripwires:
    - chocolate, tomatoes, citrus fruits
  - No significant effect:
    - dairy products (milk, cheese)



- Environmental restrictions that may trigger reaction
  - Sun exposure, strong winds, strenuous exercise, change in humidity/weather
  - Emotional factors (stress, fear, anxiety, embarrassment)
  - Heliobacter pylori (bacteria linked with stomach ulcers)
  - Vasodilators

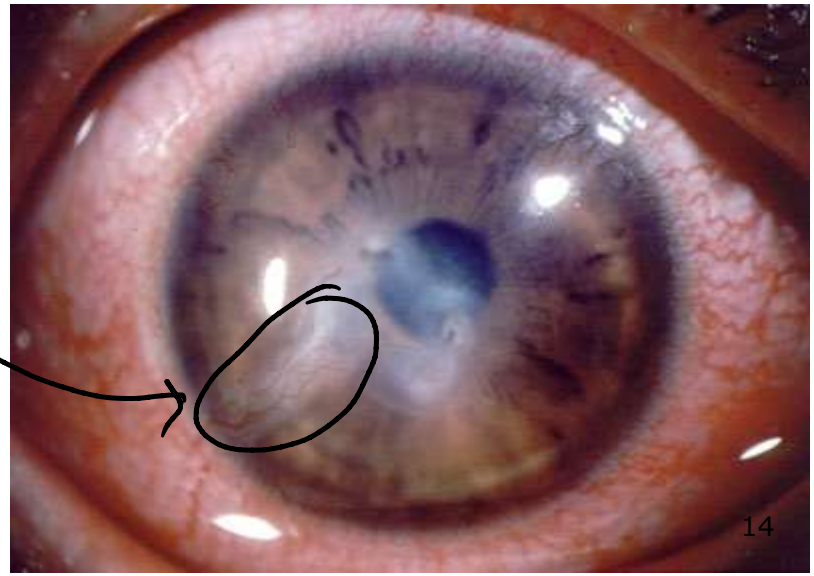
Ocular Manifestations of Rosacea

- Meibomian gland dysfunction
- Blepharitis
- Hordeolum
- Chalazion



- Inflammation and infection of the lids
- Poor tear film
- Conjunctival injection
- PEE (SPK)
- Corneal vascularization
- Corneal thinning

can cause loss of sight



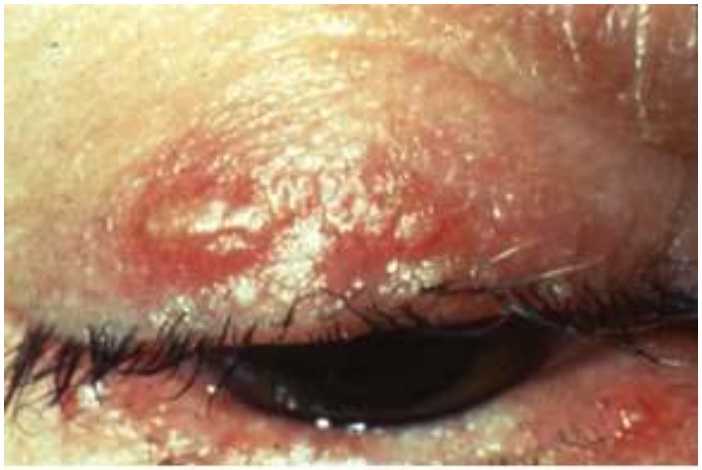
Rosacea

- Papules and pustules
- Telangiectasias
- Staph blepharitis
- Madarosis
- Meibomian gland dysfunction
- Conjunctivitis



Ocular Rosacea

- **Infection and inflammation of the lids**

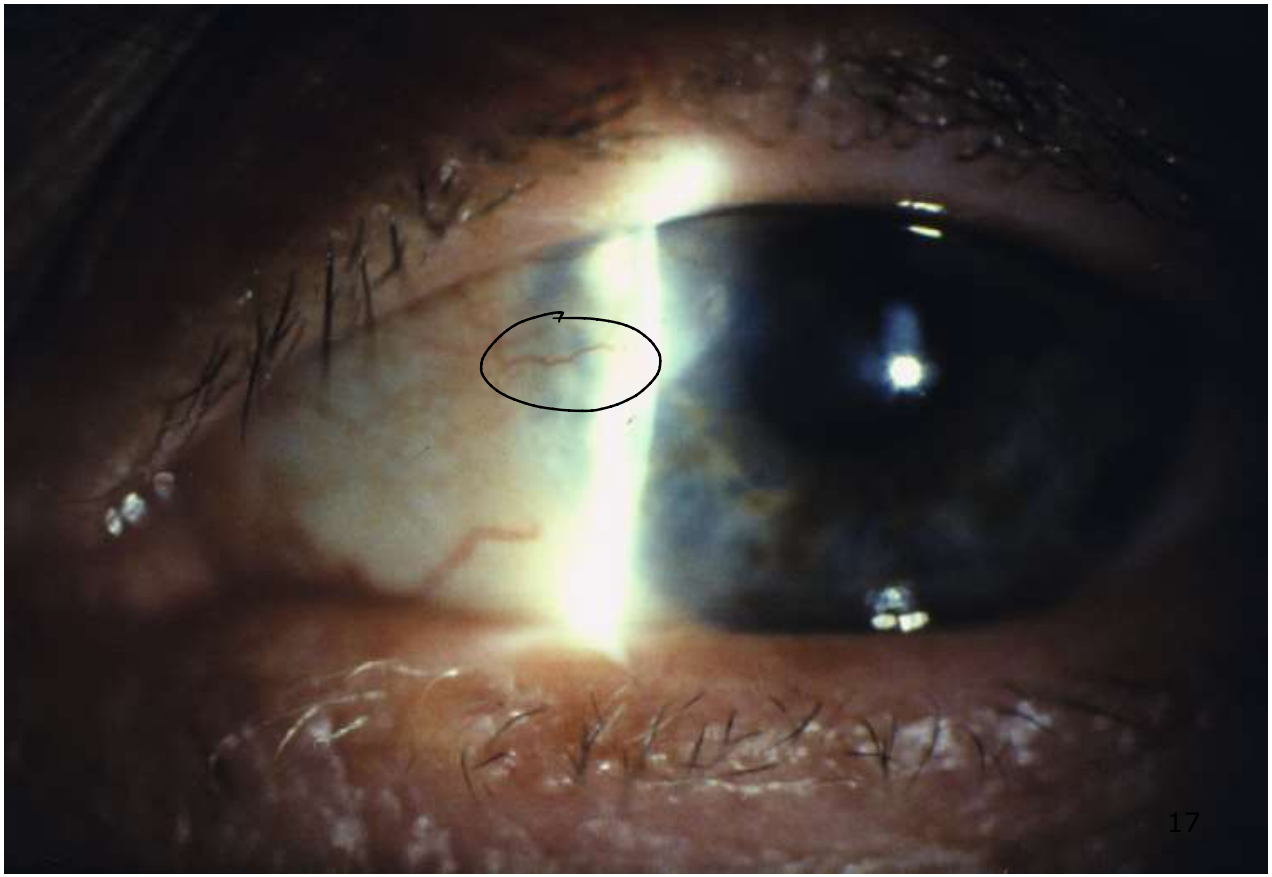




Ocular Rosacea

- **Keratitis**
  - **Corneal vascularization and thinning**

new vascolarization



Ocular Rosacea

Treatment

- **Meibomian gland dysfunction** "excessiv" heat
  - **Warm compresses, lid scrubs, gland expression**
- **Blepharitis**
  - **Same as staphylococcal blepharitis**
- **Keratitis**
  - **Artificial tears/ointments**
  - **Steroids if severe inflammation**



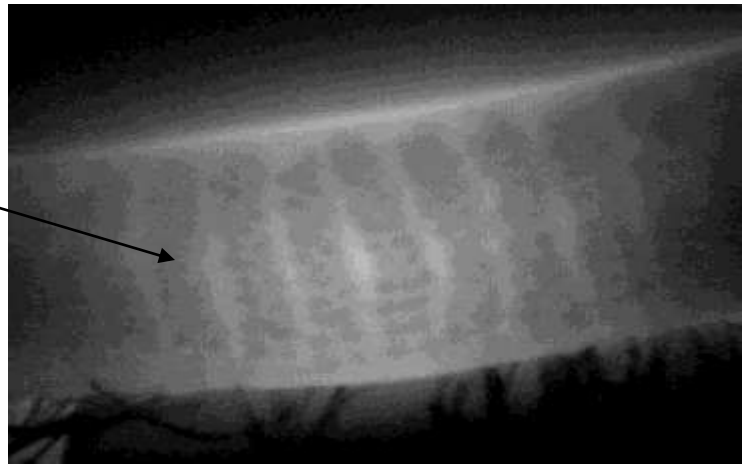
Ocular Rosacea

- Follow-up
  - Without corneal involvement - weeks
  - With corneal involvement - days
- Patient education
  - Lid hygiene
  - Dietary restrictions (triggers)
  - Environmental triggers

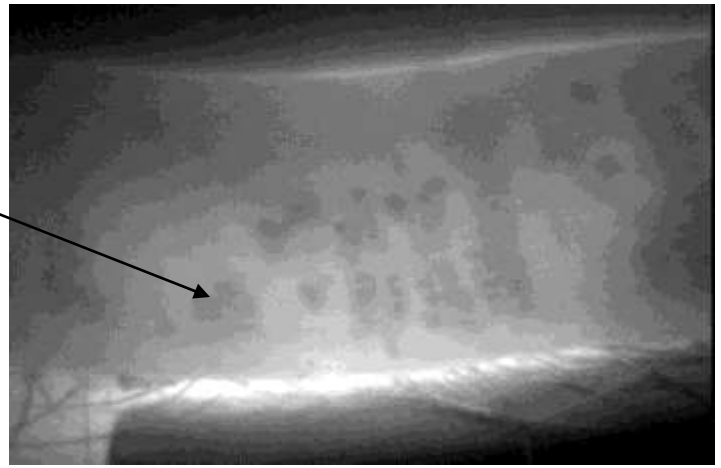


Meibomian Gland Dysfunction IR images

- **Normal meibomian glands**



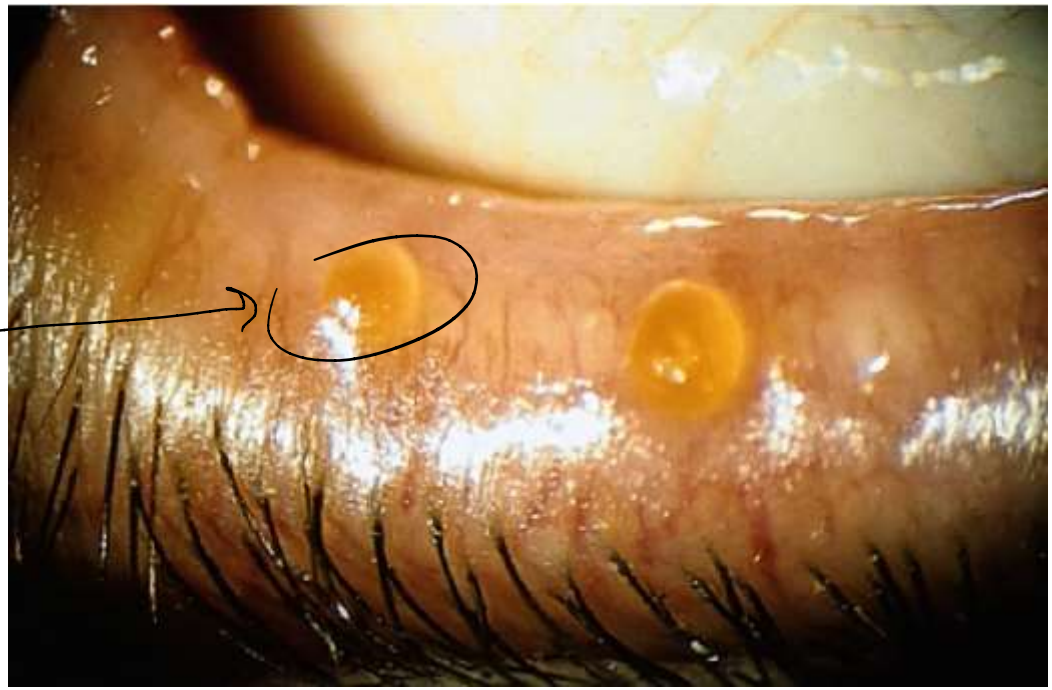
- **Abnormal meibomian glands**



Meibomian Gland Dysfunction

- **Plugged meibomian glands**
- **Meibomian 'caps'**

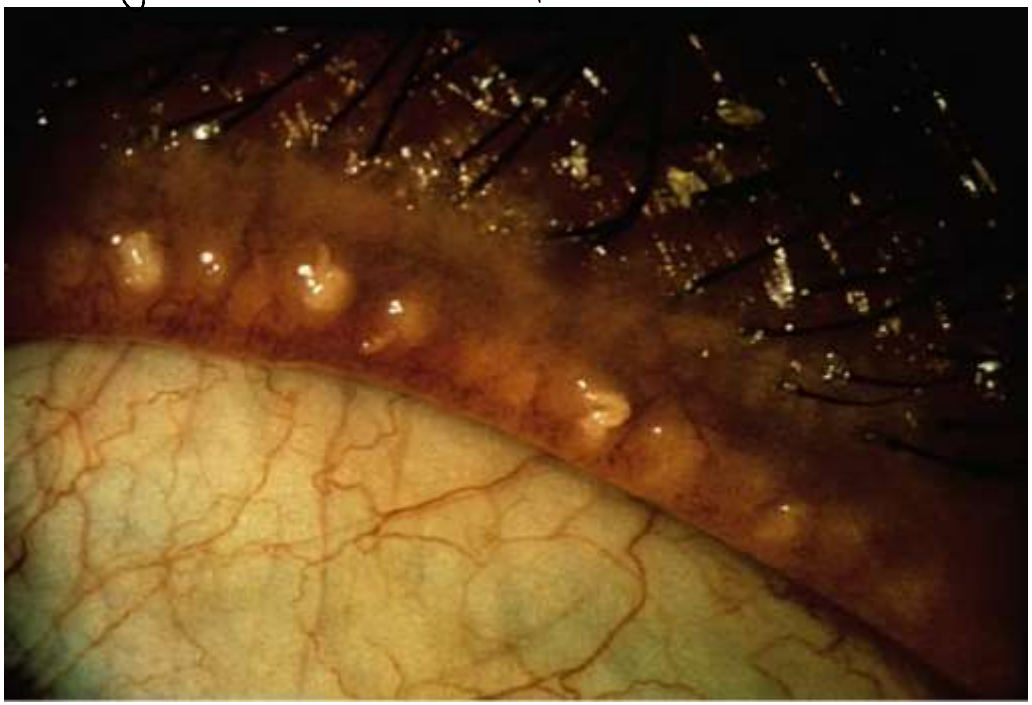
it is dry, like a glue on top when it drys out.



Meibomian Gland Dysfunction

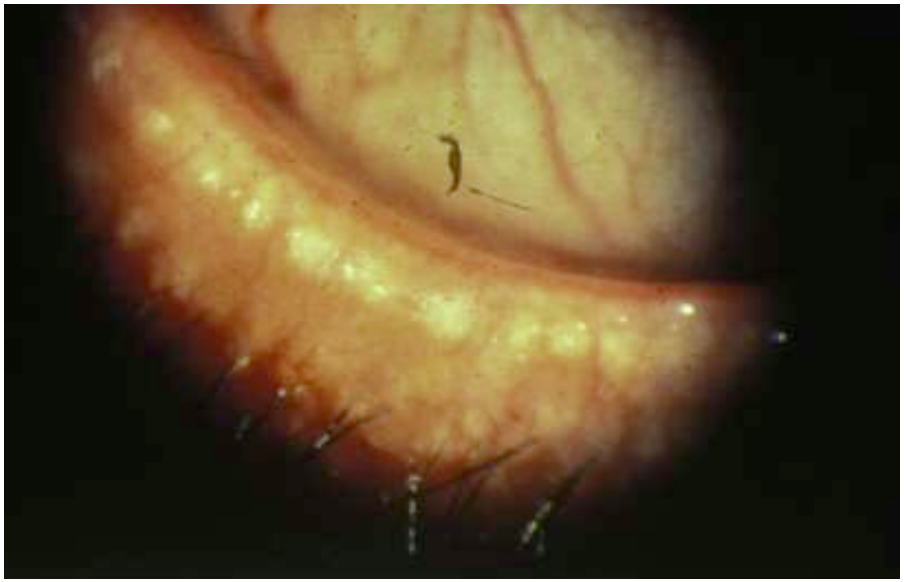
- Significant pouting of meibomian glands
- Inflammation of lid surrounding glands
- Serrated lid margin
- Tarsal conjunctival streaking
- Conjunctival injection

(Squize it out, preworm lids)



clear oily secretion not like a "toothpaste"

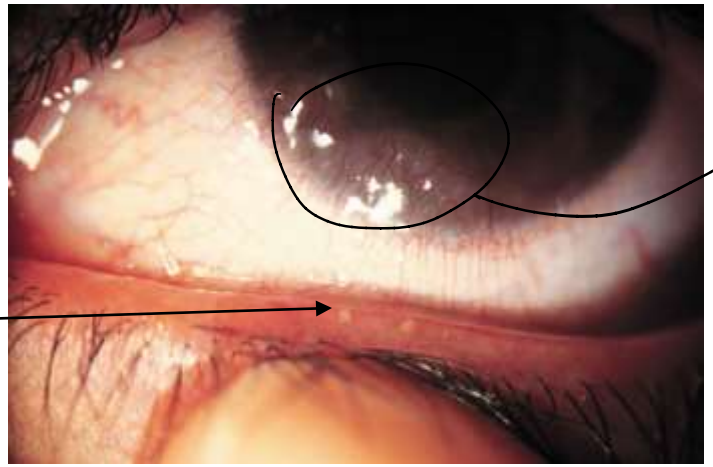
Meibomian Gland Dysfunction



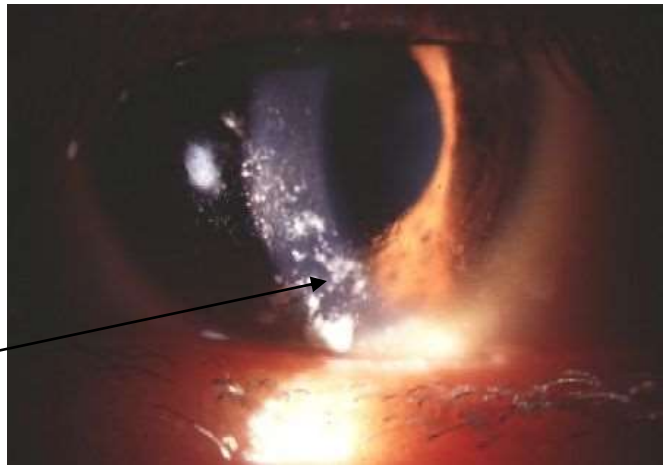


Meibomian Gland Dysfunction

- **Diagnosis usually made clinically**



- **Forced expression of the meibomian glands**
  - **"Toothpaste" test - thickened secretions are forced out of some glands** (10s wait it comes out)
  - **Many glands will yield no visible secretion**



  - **Thickened secretions in the tears**

upper lid is also good to do it, but it is more uncomfortable

Meibomian Gland Dysfunction

Treatment

- Warm compresses
  - Softens any material on the meibomian gland orifices
  - Softens thickened secretions in glands
- In-office and at-home meibomian gland expression (digital massage)
  - Pressure on the lids from the part of the tarsus away from lid margin towards lid margin (opening) - hot enough and gently!
  - Open the ducts of the glands and express the thickened secretions

Meibomian Gland Dysfunction



Titanium paddle

Meibomian Gland Dysfunction

Treatment

- **If disease severe or not responding to therapy**
  - Consider doxycycline 100 mg bid for 6 to 10 weeks
  - Taper to lowest dose or, if possible, discontinue
  - Works by inhibiting lipase production



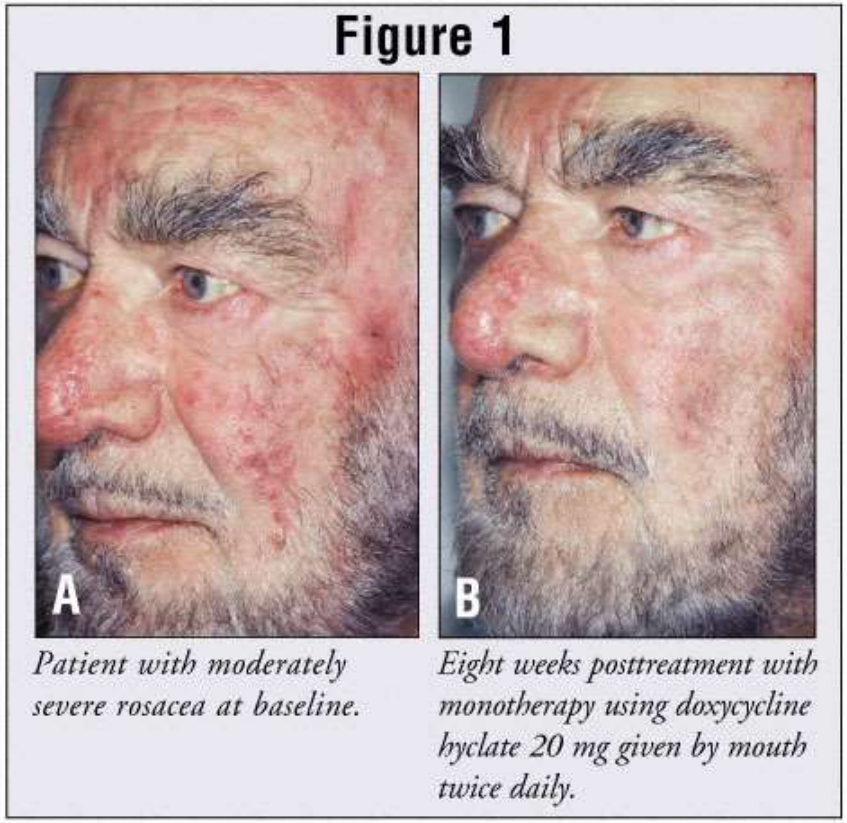
- **Manage dry eye**
- **Evaluate overlap with staphylococcal blepharitis**
  - Treat both conditions
- **Evaluate overlap with seborrheic blepharitis**
  - Treat both conditions
- **Evaluate if rosacea present**
  - Consider dermatologic consult



Meibomian Gland Dysfunction

Follow-up

- 1 week
- Then 3-4 weeks later

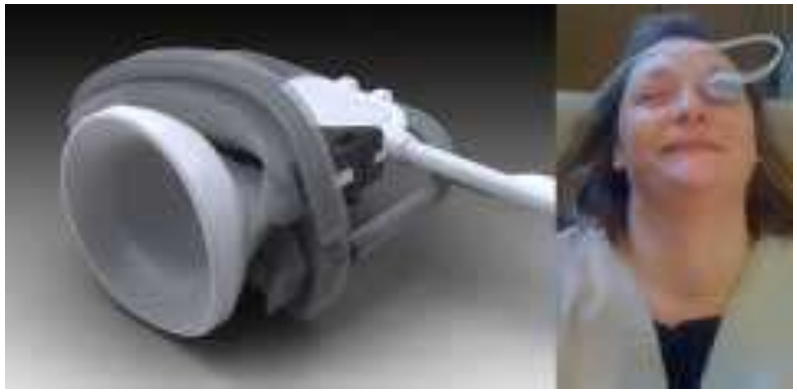


Patient education

- Chronic condition
- No cure, management only

LipiFlow Thermal Pulsation System (TearScience)

- Meibomian expression in-office treatment device
- Invented by Donald Korb, O.D.
- Applies heat to tarsal side, pressure to skin side
- Can adjust heat/pressure; 12 minute sessions
- FDA-approved 7/11/11





Intense Pulsed Light Therapy (IPL)



ful bright light. It is cold

Intense Pulsed Light Therapy (IPL)

- Non-laser, high intensity light
- Hand piece contacts skin
- Intensity, duration and wavelengths vary based on skin type/treatment goals
- Thermal heating of meibum, superficial telangiectasia



- Potential benefit for ocular rosacea and dry eye related to meibomian gland dysfunction (MGD)
- Increases lipid layer and noninvasive TBUT

Impetigo

- Skin infection with *Staphylococcus aureus* or *Streptococcus pyogenes*
- Most common
  - Children under age 6
  - Sometimes very elderly
  - Exposed to poor hygienic conditions
- Contagious



poor water conditions
poor hygienic

Impetigo

- Rash with crusts (‘honey crusts’)
- Face, distal extremities







Impetigo TREATMENT

- Oral antibiotic
  - Culture and susceptibility testing
  - Cephalosporin
  - Macrolide
  - Dicloxacillin
- Wash involved areas
- Topical antibiotic after washing
  - Bacitracin
  - Mupirocin



Erysipelas

mimic rosacea

- Skin infection involving the dermis under epidermis
  - Superficial dermal lymphatics swell up
  - local redness, heat, swelling, and highly characteristic raised, indurated border
- Most common in young and elderly
- Legs and face most frequently affected sites
- Caused by ***Streptococcus pyogenes*** also IMPETIGO is made for the same





looks like 2th degree burn

Erysipelas

- Fever, chills
- Malaise
- Painful edematous patches on skin
- If on face, can continue into lids
- Sharply demarcated areas of swelling and redness



Erysipelas







- Treatment
  - PCP and/or dermatologic consult
  - Antibiotic: Penicillin G (injected) or penicillin V (PO)

THANK YOU!